

Duality

Constraint relaxation

Michel Bierlaire

Optimization: principles and algorithms

Duality

*We take a different point of view about the optimization problem.
The point of view of the person who defines the constraint.
We introduce an important concept in optimization: duality.*

Simple example



Simple example



Simple example



Deal

- ▶ Win 1 € per meter of altitude.
- ▶ Must stay in the Alps.



Simple example



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Model

x : position, $f(x)$: altitude at x

$$\max_x f(x) \text{ s.t. } x \in \text{Alps}$$



Simple example



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- ▶ Win 1 € per meter of altitude.
- ▶ Must stay in the Alps.

Model

x : position, $f(x)$: altitude at x

$$\max_x f(x) \text{ s.t. } x \in \text{Alps}$$

Solution for the alpinist

Mont Blanc: 4807 €



Simple example



4807m

Simple example



4807m

New deal

- ▶ Pay a fine if out of the Alps.
- ▶ What fine?

Simple example



4807m



8848m

New deal

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Simple example



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8848m

New deal

- ▶ Pay a fine if out of the Alps.
- ▶ What fine?

Model

x : position, $f(x)$: altitude at x ,
 $a(x)$ =fine at x

$$\max_x f(x) - a(x)$$

Simple example



4807m



8848m

New deal

- Pay a fine if out of the Alps.
- What fine?

Model

x : position, $f(x)$: altitude at x ,
 $a(x)$ =fine at x

$$\max_x f(x) - a(x)$$

Solution

$$a(x) = 4041 \text{ € if } x \notin \text{Alps}$$

Duality

Primal problem

- ▶ Point of view of the alpinist
- ▶ Optimization under strict constraint



Dual problem

- ▶ Point of view of the billionaire
- ▶ Penalization of the constraints



Constraint relaxation

Primal problem

$$\min_{x \in \mathbb{R}^2} 2x_1 + x_2$$

subject to

$$1 - x_1 - x_2 = 0$$

$$x_1 \geq 0$$

$$x_2 \geq 0$$

Solution: $(0, 1)$

Optimal value: 1

Constraint relaxation

Primal problem

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Constraint relaxation

$$\min_{x \in \mathbb{R}^2} 2x_1 + x_2 + \lambda(1 - x_1 - x_2)$$

subject to

$$1 - x_1 - x_2 = 0$$

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Constraint relaxation

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Constraint relaxation

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Dual problem

What is the value of λ ?

$$\lambda = 0$$

$$\min_{x \in \mathbb{R}^2} 2x_1 + x_2 + \lambda(1 - x_1 - x_2) = 2x_1 + x_2$$

subject to

$$x_1 \geq 0$$

$$x_2 \geq 0$$

Solution: $(0, 0)$

Optimal value: $0 < 1$.

$$\lambda = 0$$

$$\min_{x \in \mathbb{R}^2} 2x_1 + x_2 + \lambda(1 - x_1 - x_2) = 2x_1 + x_2$$

subject to

$$x_1 \geq 0$$

$$x_2 \geq 0$$

Solution: $(0, 0)$

Optimal value: $0 < 1$.

Comments

- ▶ We obtain a lower bound,
- ▶ but a strict one.

$$\lambda = 2$$

$$\min_{x \in \mathbb{R}^2} 2x_1 + x_2 + \lambda(1 - x_1 - x_2) = 2 - x_2$$

subject to

$$x_1 \geq 0$$

$$x_2 \geq 0$$

Solution: none

Optimal value: $-\infty < 1$.

$$\lambda = 2$$

$$\min_{x \in \mathbb{R}^2} 2x_1 + x_2 + \lambda(1 - x_1 - x_2) = 2 - x_2$$

subject to

$$x_1 \geq 0$$

$$x_2 \geq 0$$

Solution: none

Optimal value: $-\infty < 1$.

Comments

- ▶ This value of λ generates an unbounded problem.
- ▶ It must be avoided.

$$\lambda = 1$$

$$\min_{x \in \mathbb{R}^2} 2x_1 + x_2 + \lambda(1 - x_1 - x_2) = x_1 + 1$$

subject to

$$x_1 \geq 0$$

$$x_2 \geq 0$$

Solution: $(0, x_2)$, $\forall x_2$

Optimal value: $1 = 1$.

$$\lambda = 1$$

$$\min_{x \in \mathbb{R}^2} 2x_1 + x_2 + \lambda(1 - x_1 - x_2) = x_1 + 1$$

subject to

$$x_1 \geq 0$$

$$x_2 \geq 0$$

Solution: $(0, x_2)$, $\forall x_2$

Optimal value: $1 = 1$.

Comments

- ▶ We obtain the same optimal value as for the primal.
- ▶ The same solution is optimal as well.

Summary

We may have a dual view on the optimization problem.

The original problem is called the primal problem. Constraints can be relaxed, with a penalty to be “paid” for their violation.

Finding the right penalty is the dual problem.